

GEN-3 SAFE HELICOPTER

DOCUMENT 00

10-TON / 7-ROTOR SAFETY-CENTERED BASELINE

Central Protected Fixed Safety Capsule Configuration

Seven equally sized 6.0 m rotors distributed around the central capsule

Four turboshaft-class prime movers • 7 mechanical damper landing legs

Conceptual Design Document • Revision 00-T10-R6-Opt

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PROPRIETARY CONCEPT INFORMATION

Concept / Pre-Feasibility Baseline — Not Certified

***Baseline lock:** This optimized revision establishes the compact 10-ton architecture with seven **6.0 m** rotors distributed around a centrally protected fixed safety capsule (no primary lift rotor directly above the capsule), **1,700 L** fuel capacity, four turboshaft-class prime movers (installed power target 2.5–2.8 MW), seven external mechanical damper landing legs, and a ten-layer safety architecture. Overall target envelope: 17 m × 17 m × 5 m.*

1. Document Control

Item	Controlled Value
Document Number	00
Revision	00-T10-R6-Opt
Project	Gen-3 Safe Helicopter
Product Baseline	10 metric tonnes / 7-rotor compact platform
Rotor Configuration	7 × 6.0 m equally sized rotors distributed around central capsule
Central roof rotor	None (no primary lift rotor directly above capsule)
Overall Envelope	17 m × 17 m × 5 m
Main body height	4 m nominal
Capsule rise above body	≈ 1 m
Fuel Planning Capacity	1,700 L nominal
Prime Movers	4 × turboshaft-class (≈ 650–700 kW class each)
Installed Power Target	2.5 – 2.8 MW
External Landing Gear	7 mechanical damper legs with thick energy-absorbing tips
Capsule Landing Support	6 mechanical damper elements (integrated with capsule isolation)
Safety Capsule	Fixed / non-jettisonable / centrally protected
Occupant Capacity	7
Founder	Amirbahador Atai
Status	Concept / pre-feasibility; engineering validation pending

This document is a design baseline, not a type certificate, flight manual, structural substantiation package, or production drawing set. All numerical values are targets or preliminary estimates until closed by analysis, supplier data and testing.

2. Executive Summary

This optimized baseline deliberately combines the strongest elements of previous revisions: a compact 10-ton demonstrator scale, seven equally sized 6.0 m rotors arranged around a centrally protected fixed safety capsule, removal of any primary lift rotor from directly above the occupant volume, seven external mechanical damper landing legs, and a clear ten-layer safety architecture.

- 10,000 kg maximum design mass target.
- Seven primary lift rotors, each 6.0 m diameter, distributed around the central capsule (no rotor on the central roof line above the capsule).
- Total geometric disk area $\approx 197.9 \text{ m}^2$; disk loading $\approx 50.5 \text{ kg/m}^2$ at 10 t.
- 1,700 L nominal fuel capacity ($\approx 1,360 \text{ kg}$ at 0.80 kg/L).
- Four turboshaft-class prime movers, installed power target 2.5–2.8 MW.
- Overall envelope $17 \text{ m} \times 17 \text{ m} \times 5 \text{ m}$ (main body $\approx 4 \text{ m}$; capsule rises $\approx 1 \text{ m}$ above main body).
- Central fixed safety capsule with multi-angle carbon-fiber roll-cage, high-stroke mechanical isolation, and soft damper-mounted seats.
- Seven external mechanical damper landing legs with progressive energy absorption and thick soft tip sections.
- Six capsule-side mechanical damper elements forming part of the isolation chain.
- Aircraft-level emergency recovery concept (8 advanced parachutes).
- Fire detection, isolation and suppression as independent safety functions.
- Architecture remains modular for a future detachable-capsule derivative (Variant B).

The central engineering proposition remains layered survivability. The aircraft does not depend on a single rescue mechanism. It combines structural containment, mechanical isolation, multiple damped load paths, fire protection, distributed propulsion, fault reconfiguration, emergency recovery concepts, and multi-stage occupant energy management.

3. Frozen / Controlled Product Baseline

Parameter	Baseline	Status
Maximum design mass	10,000 kg target	Controlled target
Rotor count	7	Controlled
Rotor diameter	6.0 m nominal (all equal)	Controlled
Rotor arrangement	Distributed around central capsule; no central-roof rotor	Controlled
Total geometric disk area	$\approx 197.9 \text{ m}^2$	Calculated
Disk loading at 10 t	$\approx 50.5 \text{ kg/m}^2$	Calculated
Overall envelope	17 m $\times$ 17 m $\times$ 5 m	Controlled target
Main body height	4 m nominal	Controlled target
Capsule rise above body	$\approx 1 \text{ m}$	Controlled target
Fuel capacity	1,700 L nominal	Controlled target
Fuel mass (reference)	$\approx 1,360 \text{ kg}$ at 0.80 kg/L	Planning value
Prime movers	4 $\times$ turboshaft-class	Controlled direction
Power class per unit	$\approx 650\text{--}700 \text{ kW}$ class	Target class
Installed power target	2.5 – 2.8 MW	Controlled target
Approx. real hover power (FM ≈ 0.72)	$\approx 1.94 \text{ MW}$	Calculated with assumptions
External landing legs	7 mechanical damper legs	Controlled
Capsule damper elements	6 mechanical damper elements	Controlled
Safety capsule	Fixed / non-jettisonable / centrally protected	Controlled
Primary safety philosophy	Ten-layer containment, isolation, energy absorption, fault tolerance, emergency recovery	Controlled

4. Vehicle Architecture

The platform is organized around a centrally located, protected safety capsule. Seven equally sized 6.0 m rotors are distributed around the capsule so that no primary lift rotor is positioned directly above the occupant volume. This arrangement improves occupant protection from rotor debris, fire and mechanical failure while preserving distributed lift and control authority.

Four turboshaft-class prime movers feed a segmented electrical distribution system supplying seven independent rotor drive channels. Any single prime mover or any single rotor channel can be isolated without immediate total loss of lift authority.

- Central occupant capsule positioned near the vehicle center of gravity and raised ≈ 1 m above the main body line.
- Primary airframe forms a load-carrying perimeter around the capsule.
- Seven rotor stations distributed around the envelope for lift redundancy and control authority.
- Fuel and hot machinery separated from the occupied capsule by dedicated containment zones.
- Seven external mechanical damper landing legs route ground-impact loads through energy-absorbing paths before they reach the capsule.
- Six additional capsule-side damper elements form part of the multi-stage isolation chain.
- Architecture prioritizes controlled graceful degradation rather than reliance on one heroic emergency device.

5. Ten-Layer Safety Architecture

The ten-layer architecture is the core product identity. Layers are overlapping and designed to be independently reviewable and testable.

Layer	Function	Implementation Concept
1	Occupant Survival Capsule	Central fixed capsule with thick multi-angle carbon-fiber roll-cage
2	Multi-stage Energy Absorption	Landing-leg response → airframe controlled deformation → capsule-mount stroke → roll-cage → soft seats
3	Capsule-to-Airframe Isolation	Multiple high-stroke mechanical damper mounts + 6 capsule-side damper elements
4	Distributed Propulsion	7 independent rotor channels; loss of one channel does not remove all lift
5	Prime-Mover Redundancy	4 independent turboshaft units; any single unit can be isolated
6	Fire Detection / Isolation / Suppression	Zone detection, fuel/electrical isolation, suppression around capsule perimeter and critical zones
7	Control Redundancy	Segregated sensing, computation and actuator paths with fault-containment logic
8	Emergency Descent / Recovery	8 advanced parachutes (aircraft-level) + controlled degraded flight modes
9	Water Survival	Buoyancy provisions, flotation concept, multi-orientation egress
10	Structural Containment & Modular Isolation	High-energy hazards kept outside occupant volume; modular isolation of a failed/burning rotor

No under-capsule fans are included in this fixed-capsule baseline. Safety claims remain conceptual until supported by analysis and test evidence. No layer is treated as a guaranteed survival device at this stage.

6. Fixed Safety Capsule – Detailed Features

Feature	Specification
Location	Central heart of the vehicle, raised ≈ 1 m above main body
Primary volume shape	Near-spherical / highly rounded for multi-direction impact resistance
Internal reinforcement	Thick multi-angle carbon-fiber roll-cage connecting body panels from all sides
Occupant seats	7 very soft bed-like seats with high-stroke mechanical dampers to the capsule floor
Capsule-to-airframe interface	Multiple high-stroke mechanical damper mounts
Capsule-side damper elements	6 mechanical damper elements forming part of the isolation chain
External landing legs	7 mechanical damper legs with progressive energy absorption and thick soft plastic tip sections
Fire protection around capsule	Suppression nozzles at perimeter
Egress concept	Multiple exit routes; operable after normal electrical power loss (design objective)
Parachute system	8 advanced canopies sized for whole-aircraft emergency recovery (concept)
Independent capsule flight	None (fixed non-jettisonable configuration)

Conceptual energy-absorption sequence: (A) Seven external landing-leg response → (B) Primary airframe controlled deformation → (C) Capsule support stroke and 6 damper elements → (D) Capsule shell and roll-cage load distribution → (E) Soft seat and restraint attenuation. The capsule is not described as crash-proof; the engineering target is a defined survivability envelope.

7. Seven-Rotor Propulsion Architecture (6.0 m, Distributed)

All seven rotors are equal in diameter (6.0 m) and are distributed around the central capsule. No primary lift rotor is located on the central roof line directly above the occupant volume. This improves protection of the capsule from rotor debris, fire and mechanical failure while retaining distributed lift and control authority.

Item	Preliminary value
Rotor count	7 (all equal)
Rotor diameter	6.0 m
Arrangement	Distributed around central capsule; no central-roof rotor
Area per rotor	28.27 m²
Total geometric disk area	≈ 197.9 m²
Disk loading at 10,000 kg	≈ 50.5 kg/m²
Hover thrust per rotor at 10 t	≈ 14.01 kN
Illustrative 20% total-thrust margin	≈ 16.81 kN per rotor
Ideal induced hover power (10 t, ISA SL)	≈ 1.40 MW
Approximate real power (FM ≈ 0.72)	≈ 1.94 MW

Real installed power must include profile, mechanical, electrical, control, atmospheric and reserve margins. The installed power target of 2.5–2.8 MW is intended to provide approximately 25–40% system margin over the approximate real hover power estimate.

8. Propulsion and Engine Architecture (Closed Direction)

Item	Target / Concept
Number of prime movers	4
Power class per unit	Approximately 650–700 kW class (final OEM selection pending)
Total installed power target	2.5 – 2.8 MW
Analytical rationale	Real hover power estimate ≈ 1.94 MW at 10 t; installed target provides ~25–40% system margin
Benchmark engine class	PT6C-family or equivalent turboshaft in the 650–1,000+ kW published class (installation derate applies)
Power flow	Turboshaft → generator → segmented HV bus → independent inverters → rotor motors
Key safety rule	Redundancy must be physically independent; shared vulnerable buses do not equal independent channels
Degraded operation objective	Continue controlled flight / landing after loss of any single prime mover or any single rotor channel

Final engine selection, generator matching, installation rights, export controls and certification pathway require direct OEM engagement. Published engine figures are engine-level only; installed aircraft power will be lower after conversion losses, thermal limits and accessories.

9. Fuel System, Endurance and Range

Parameter	Value
Nominal tank capacity	1,700 L
Reference fuel mass	≈ 1,360 kg (at 0.80 kg/L Jet-A)
Approximate real hover power at 10 t	≈ 1.94 MW
Hover time at 10 t (SFC ~0.32–0.35 kg/kWh, typical)	Approximately 115–130 minutes
Cruise endurance (typical mission profile)	Approximately 3.2–3.6 hours
Approximate range at 200 km/h (with normal reserves)	Approximately 620–720 km

These endurance and range figures are preliminary order-of-magnitude estimates. They depend on actual installed efficiency, mission profile, reserves policy and atmospheric conditions. They are not certified performance guarantees.

10. Preliminary Mass Budget

Mass group	Planning mass	Comment
Primary airframe and rotor support structure	2,050 kg	Includes major load paths and rotor support
Safety capsule and occupant protection structure	1,150 kg	Shell, roll-cage, isolators, dampers, seats
Seven rotors, hubs and pitch/control hardware (6.0 m)	720 kg	Supplier-dependent
Powerplant / generators / motors / drives (4 units)	1,550 kg	Four turboshaft-class + electrical chain
Fuel system excluding fuel	230 kg	Tanks, pumps, plumbing, protection
Fuel, 1,700 L planning capacity	1,360 kg	Planning density 0.80 kg/L
Avionics, flight control and electrical distribution	520 kg	Includes redundancy allowance
Landing gear / ground interface (7 external + 6 capsule)	450 kg	Updated for 7 external legs
Fire, emergency recovery and safety equipment	540 kg	Includes 8-parachute concept allocation
Thermal management / cooling	350 kg	Powerplant-dependent
Interconnects, hardware, fluids and growth reserve	1,180 kg	Development margin
Total planning mass	≈ 9,800 kg	Leaves ~200 kg to the 10,000 kg target

The residual is a design-closure reserve until propulsion and structure selections are validated with supplier data. Continuous bottom-up mass-statement discipline is mandatory.

11. Flight-Control and Fault-Reconfiguration Architecture

The seven-rotor distributed layout supports graceful degradation. The flight-control system must detect abnormal rotor speed, thrust, temperature, vibration, power and actuator behavior and reconfigure the control law within validated limits. Independent monitoring of each rotor channel, automatic isolation of a failed channel, reallocation of authority across healthy rotors, and independent emergency power for flight-critical functions are design objectives. Fault reconfiguration must not create a new unstable condition.

12. Structural and Load-Path Philosophy

The primary structure is organized into multiple load paths around the central capsule. Rotor loads, engine loads, landing loads and emergency loads are not assumed to follow the same path. Seven external mechanical damper legs and six capsule-side damper elements form the primary ground-impact and isolation load paths. Structural sizing will use limit and ultimate load cases. This document does not claim compliance with any certification basis.

13. Emergency Recovery Concept

Scenario	Primary response	Secondary response
Single rotor failure	Distributed control reconfiguration	Power redistribution / landing mode
Multiple rotor degradation	Emergency flight mode	Aircraft recovery system (8 parachutes)
Major fire	Detection, isolation, power reduction	Emergency recovery / landing
Severe structural event	Maintain capsule integrity	Aircraft recovery concept
Water impact / ditching	Capsule survivability and egress	Recovery / flotation provisions

14. Fire and Thermal Safety

Thermal sensors in engine, generator, fuel and high-energy zones; fire detection with independent logic where practical; fuel isolation valves; fire-resistant barriers between hot machinery and occupied space; suppression system sized after hazard analysis (capsule perimeter + critical zones); emergency power reduction and load shedding; post-fire safe-state logic. The safety philosophy is containment plus isolation.

15. Water Survival and Ditching

The capsule is intended to provide a protected survival volume during a water-landing scenario: structural protection, controlled flotation, protected egress, and preservation of critical emergency equipment. Water-impact survivability is a major validation item and cannot be established from conceptual geometry alone.

16. Manufacturing and Prototyping Strategy

Phase 1: coupon and material tests. Phase 2: subscale rotor, motor, controller and thermal tests. Phase 3: full-scale capsule impact and damper test articles. Phase 4: integrated propulsion test rig. Phase 5: unmanned integrated demonstrator. Phase 6: progressive flight-envelope expansion. Phase 7: certification-oriented conformity and compliance programme. The compact 10-ton baseline is intentionally chosen to enable staged physical development.

17. Verification and Validation Plan

Subsystem	Analysis	Bench / Dynamic	Flight
Rotor / hub	Yes	Yes	Yes
Propulsion channel	Yes	Yes	Yes
Flight controls	Yes	HIL + fault injection	Yes
Capsule structure	FEA	Full-scale impact	Limited
Dampers (7+6)	Model + characterization	Full-scale	Integrated
Fire protection	Hazard analysis	System fire test	As required
Emergency recovery	Simulation	Deployment test	Flight demo
Water survival	Simulation	Water-impact test	As required

18. Certification Strategy

The certification pathway is not fixed. Final category, operational concept, occupant capacity, MTOM and jurisdiction will determine the applicable basis (CS-27 / CS-29 or equivalent). This document does not claim compliance and must not be presented as evidence of compliance.

19. Safety Assessment Framework

Functional Hazard Assessment, Preliminary System Safety Assessment, common-cause/common-mode analysis, failure propagation analysis, structural failure and load-path assessment, human factors and emergency egress assessment, fault detection/annunciation assessment, and verification traceability are all required. The ten-layer architecture is not ten independent guarantees; common-mode vulnerabilities must be explicitly analyzed.

20. Product and IP Positioning

The strongest differentiating theme is an integrated architecture in which the centrally protected capsule, structural isolation, multi-stage energy absorption, distributed propulsion (no rotor above the capsule), fire containment and emergency recovery are designed as one safety system. Core assets remain candidate IP / know-how until formal legal assessment is performed.

21. Preliminary Risk Register

Risk	Impact	Current treatment	Next closure action
Mass growth above 10 t	High	Growth reserve retained	Supplier quotes + weighed hardware
Installed power higher than target	High	2.5–2.8 MW target with margin	Rotor CFD + system efficiency model
Damper stroke insufficient	High	7 external + 6 capsule elements	Dynamic impact testing
Control difficulty without central rotor	Medium-High	Distributed arrangement selected	Control-law simulation + HIL
Common-mode control failure	High	Distributed architecture	SSA + HIL fault injection
Fire propagation	High	Containment + isolation layers	Fire hazard analysis + tests
Emergency recovery load exceedance	High	8-parachute concept retained	Deployment simulation + structural test
Water-impact instability	High	Capsule survival concept	Water-impact modelling + test
Engine/generator integration mismatch	High	Early OEM engagement required	Matching studies + supplier data

22. Open Engineering Items Before Rev 01

- Freeze detailed mission profile (payload, crew, cruise speed, altitude, range, endurance).
- Complete rotor aerodynamic model (blade count, solidity, tip speed, control method) for the distributed 6.0 m arrangement.
- Final OEM selection and generator matching for the four turboshaft-class units.
- Close full power budget including realistic hover, climb and degraded-mode margins.
- Develop first-order thermal model for generators, motors and inverters.
- Replace planning mass allocations with component-level supplier data.
- Develop capsule structural load cases and damper stroke/force targets (7+6 elements).
- Define exact parachute sizing, deployment envelope and structural attachment loads.
- Quantify water-impact and flotation requirements.
- Create formal system safety architecture and failure-mode database.
- Map the final design to the applicable certification basis.
- Validate control authority and handling qualities without a central-roof rotor.

23. Design Rules for the 10-Ton Baseline

Rule 1 — Do not allow the gross-mass target to become a paper-only number.

Rule 2 — Do not claim safety credit for an emergency device before evidence exists.

Rule 3 — Prefer passive mechanical safety layers where they reduce dependence on software or electrical power.

Rule 4 — Avoid single common-mode dependencies that can defeat multiple safety layers simultaneously.

Rule 5 — Treat the capsule as a system, not simply as a stronger cabin.

Rule 6 — Keep the first demonstrator free of detachable-capsule complexity.

Rule 7 — Maintain explicit separation between controlled baseline values, engineering targets, calculated estimates, and demonstrated performance.

24. Preliminary Performance Envelope

Metric	Current treatment
Gross mass	10,000 kg target
Hover thrust requirement	≈ 98.1 kN total; ≈ 14.01 kN per rotor
Illustrative 20% total thrust margin	≈ 117.7 kN total; ≈ 16.81 kN per rotor
Ideal induced hover power floor	≈ 1.40 MW at ISA sea-level density
Approximate real hover power (FM ≈ 0.72)	≈ 1.94 MW
Installed power target	2.5 – 2.8 MW
Hover endurance (1,700 L, typical SFC)	Approximately 115–130 minutes
Cruise endurance (typical)	Approximately 3.2–3.6 hours
Approximate range at 200 km/h	Approximately 620–720 km (with reserves)
Cruise speed	TBD after detailed aerodynamic model

The ideal and approximate real hover figures are sanity checks only. They are not claims of achievable certified aircraft performance.

25. Investor and Due-Diligence Readiness

The optimized 10-ton baseline with distributed 6.0 m rotors (no rotor above the capsule), seven external mechanical damper legs, closed propulsion direction (4 turboshaft-class units, 2.5–2.8 MW), 1,700 L fuel and ten-layer safety architecture is intended to be easier to communicate and demonstrate than earlier very-large concepts. A professional investor package should pair this document with a one-page product sheet, a mass/power spreadsheet, a preliminary hazard analysis, a test roadmap, and evidence from physical demonstrators.

26. Glossary

Term	Meaning
CDD	Conceptual Design Document
FM	Figure of Merit (rotor efficiency metric)
HIL	Hardware-in-the-loop
FEA	Finite element analysis
MTOM	Maximum take-off mass
FHA	Functional Hazard Assessment
SSA	System Safety Assessment
Fixed capsule	Safety capsule that remains mechanically integrated and is not designed to separate
Graceful degradation	Controlled reduction in capability after defined failures while preserving required functions
Installed power	Total shaft/electrical power available at aircraft level after installation losses

27. Final Baseline Statement

Document 00 (Revision 00-T10-R6-Opt) establishes the optimized Gen-3 compact flagship baseline: a 10-ton rotorcraft with seven equally sized **6.0 m** rotors distributed around a centrally protected fixed safety capsule (no primary lift rotor directly above the capsule), **1,700 L** fuel capacity, four turboshaft-class prime movers with an installed power target of **2.5–2.8 MW**, seven external mechanical damper landing legs, six capsule-side damper elements, an overall envelope of 17 m × 17 m × 5 m, and a ten-layer safety architecture.

The central engineering proposition is layered survivability. The aircraft does not depend on one rescue mechanism. It combines structural containment, mechanical isolation, multiple damped load paths, fire protection, distributed propulsion, fault reconfiguration, emergency recovery concepts, and multi-stage occupant energy management.

The baseline remains deliberately conservative about claims. Quantitative performance, mass, power, recovery performance and certification compliance remain subject to analysis and test. The next revision should close the detailed rotor aerodynamic model for the distributed arrangement, OEM engine/generator package, and first component-level mass statement.

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Document: 00 | Revision: 00-T10-R6-Opt | Status: Concept / Pre-Feasibility (Optimized)

END OF DOCUMENT 00 — OPTIMIZED BASELINE